

1. A half-wave rectifier with a Si diode having a cut-in voltage of 0.7 V is used to supply 50 V DC to a resistance load of $800\ \Omega$. Calculate the peak value of a sinusoidal AC input required for the same

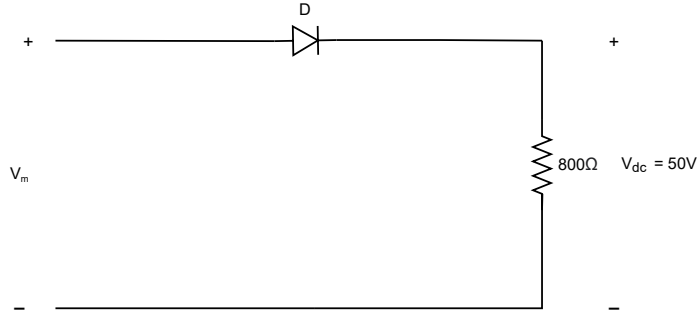


Figure 1

Solution:

The given circuit is redrawn in Figure 2.

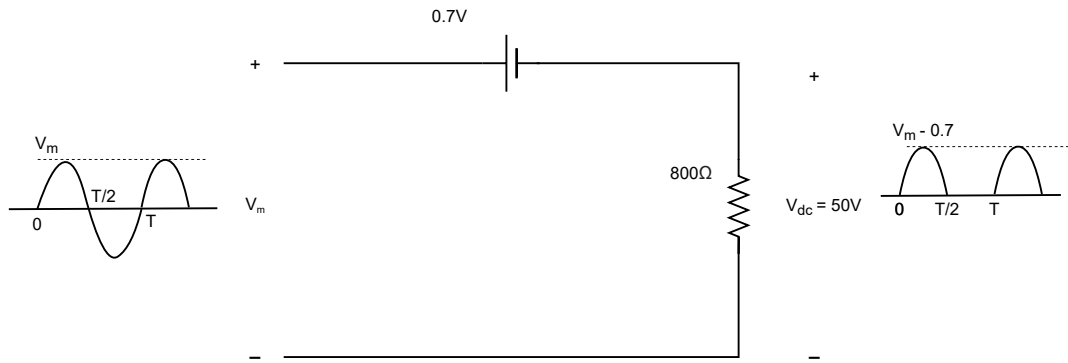


Figure 2

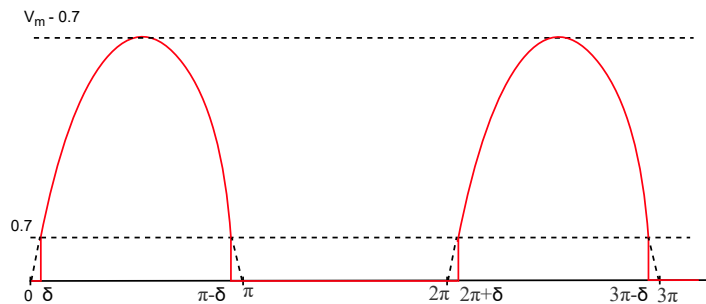


Figure 3

Referring to Fig.3, the output waveform, we can say that the diode conducts between angle $\delta \leq \theta \leq \pi - \delta$, where $\theta = \omega t$ and $\delta = \sin^{-1}(\frac{0.7}{V_m})$.

The DC voltage V_{DC} is given as

$$V_{DC} = \frac{1}{2\pi} \left(\int_{\delta}^{\pi - \delta} (V_m \sin \theta - 0.7) d\theta + \int_{\pi}^{2\pi} 0 \sin \theta d\theta \right)$$

$$= \frac{1}{2\pi} \int_{\delta}^{\pi-\delta} V_m \sin \theta \, d\theta - \frac{1}{2\pi} \int_{\delta}^{\pi-\delta} 0.7 \, d\theta$$

Approximating $\delta = \sin^{-1}(\frac{0.7}{V_m})$ to zero, we get

$$\begin{aligned} V_{DC} &\approx \frac{1}{2\pi} \int_0^{\pi} V_m \sin \theta \, d\theta - \frac{1}{2\pi} \int_0^{\pi} 0.7 \, d\theta \\ &\approx \frac{V_m}{\pi} - \frac{0.7}{2} \end{aligned}$$

Hence,

$$V_m \approx 158.17 \, V$$

2. Plot the voltage across the load R for the given input. Assume that the time constant RC of the circuit is large compared to T , the time period of the signal. The diodes are considered ideal.

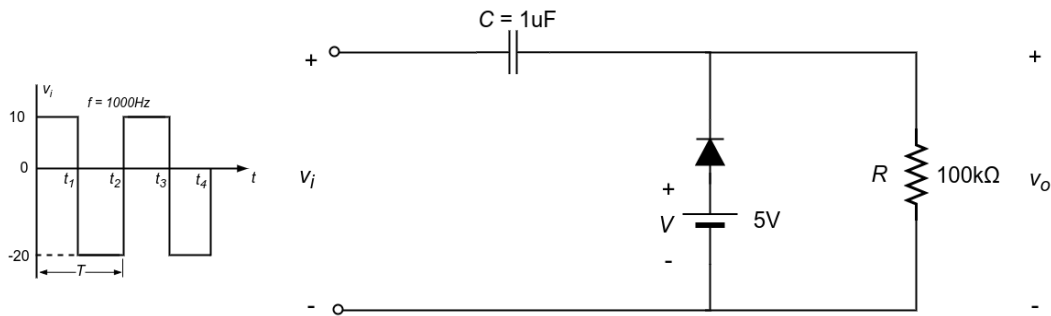


Figure 4

Solution:

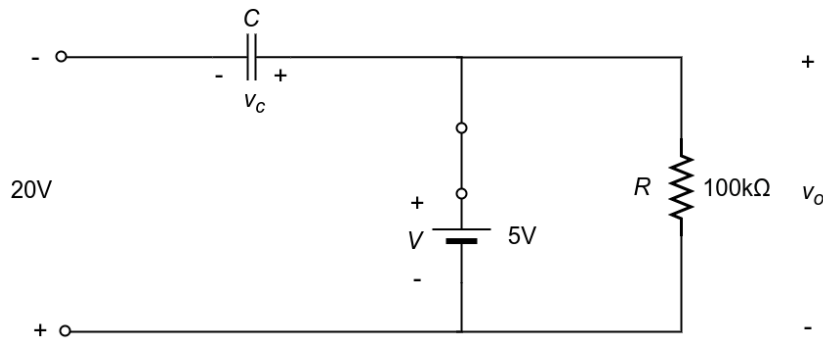


Figure 5

In general, we can apply KVL when diode is ON and write: $-V_i + 5 - V_C = 0$

In the negative half-cycle, the diode is on as shown in Fig. 5 with the input voltage $V_i = -20 \, V$.

The voltage across the capacitor is given by

$$20 + 5 = V_C$$

$$V_C = 25 \, V$$

We can now write

$$v_o = V_i + 25$$

For $V_i = -20V$, we have

$$v_o = 5V$$

In the positive half-cycle, the diode is off as shown in Fig. 6. As the time constant RC is larger than T , hence

$$v_o = 25V + 10V = 35V$$

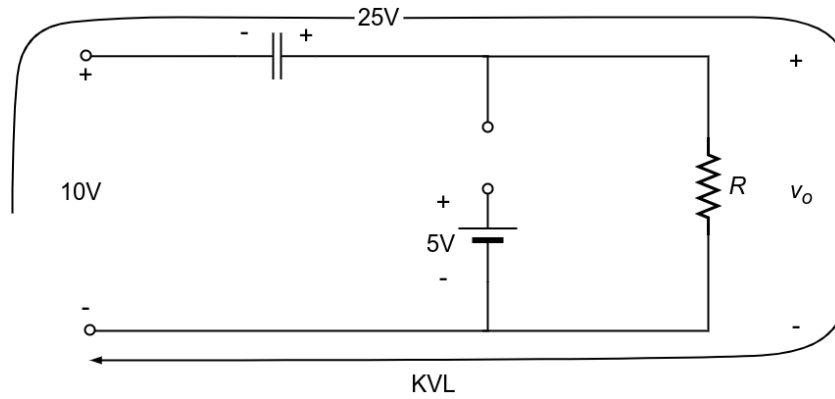


Figure 6

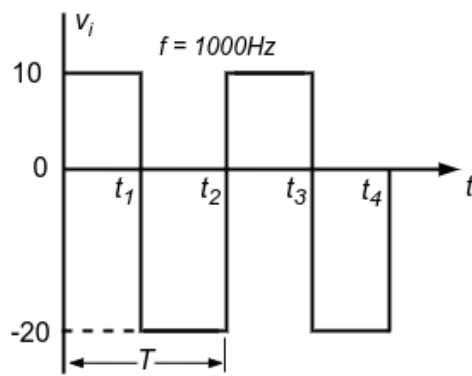


Figure 7 Input waveform

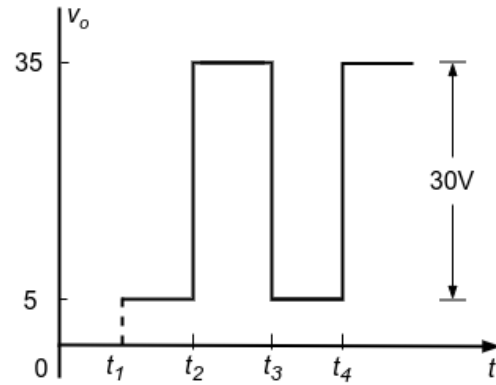


Figure 8 Output waveform

3. For the Zener diode network, determine V_L , V_R , I_Z and P_Z for (a) $R_L = 1.2k\Omega$ and (b) $R_L = 3k\Omega$

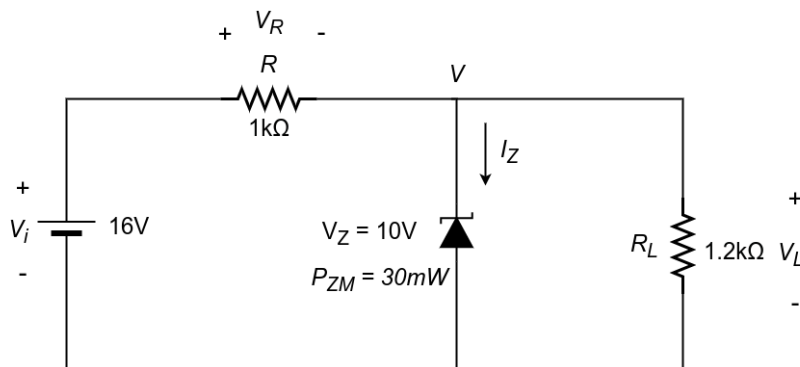


Figure 9

Solution:

(a) For $R_L = 1.2k\Omega$

We have

$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2k\Omega(16V)}{1k\Omega + 1.2k\Omega} = 8.73V$$

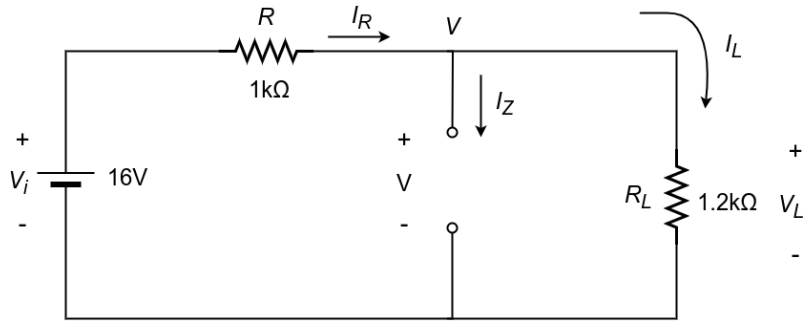


Figure 10

Since $V = 8.73V < V_Z = 10V$, the diode is in an “off” state. Hence

$$\begin{aligned} V_L &= V = 8.73V \\ V_R &= V_i - V_L = 16V - 8.73V = 7.27V \\ I_Z &= 0A \end{aligned}$$

The power dissipated is

$$P_Z = V_Z I_Z = V_Z (0A) = 0W$$

(b) For $R_L = 3k\Omega$

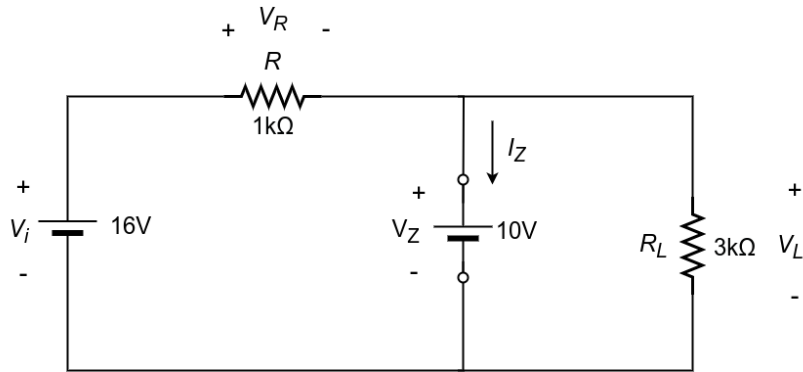


Figure 11

We have

$$V = \frac{R_L V_i}{R + R_L} = \frac{3k\Omega(16V)}{1k\Omega + 3k\Omega} = 12V$$

Since $V = 12V > V_Z = 10V$, the diode is in an “on” state. Hence

$$\begin{aligned} V_L &= V = 10V \\ V_R &= V_i - V_L = 16V - 10V = 6V \\ I_L &= \frac{V_L}{R_L} = \frac{10V}{3k\Omega} = 3.33mA \\ I_R &= \frac{V_R}{R} = \frac{6V}{1k\Omega} = 6mA \\ I_Z &= I_R - I_L = 6mA - 3.33mA = 2.67mA \end{aligned}$$

The power dissipated is

$$P_Z = V_Z I_Z = 10V(2.67mA) = 26.7mW$$

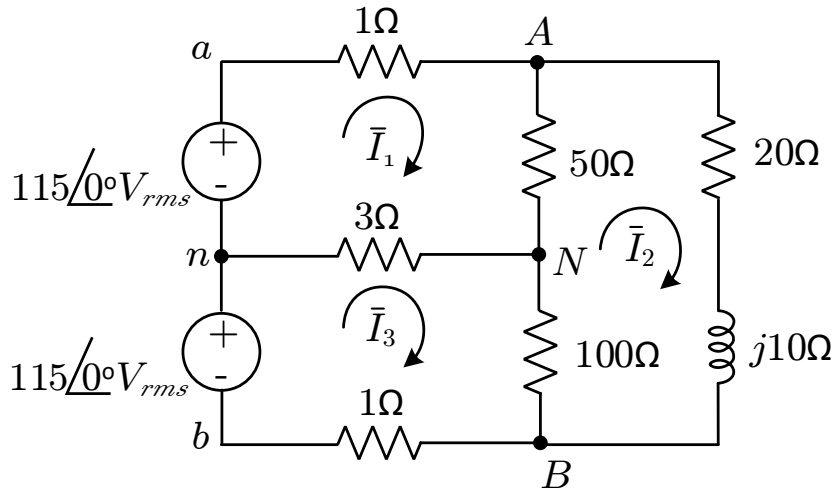


Figure 12 A typical single-phase three-wire system.

4. Analyze the system shown in the Figure below and determine the power delivered to each of the three loads as well as the power lost in the neutral wire and each of the two lines.

Note: $\bar{b}$ indicates phasor quantity.

Solution:

The three mesh equations are:

$$\begin{aligned} -115\angle 0^\circ + \bar{I}_1 + 50(\bar{I}_1 - \bar{I}_2) + 3(\bar{I}_1 - \bar{I}_3) &= 0 \\ (20 + j10)\bar{I}_2 + 100(\bar{I}_2 - \bar{I}_3) + 50(\bar{I}_2 - \bar{I}_1) &= 0 \\ -115\angle 0^\circ + 3(\bar{I}_3 - \bar{I}_1) + 100(\bar{I}_3 - \bar{I}_2) + \bar{I}_3 &= 0 \end{aligned}$$

The equations can be simplified to:

$$\begin{aligned} 54\bar{I}_1 - 50\bar{I}_2 - 3\bar{I}_3 &= 115\angle 0^\circ \\ -50\bar{I}_1 + (170 + j10)\bar{I}_2 - 100\bar{I}_3 &= 0 \\ -3\bar{I}_1 - 100\bar{I}_2 + 104\bar{I}_3 &= 115\angle 0^\circ \end{aligned}$$

Solving for the phasor currents $\bar{I}_1$, $\bar{I}_2$, and $\bar{I}_3$, we find

$$\begin{aligned} \bar{I}_1 &= 11.24\angle -19.83^\circ \text{ A} \\ \bar{I}_2 &= 9.389\angle -24.47^\circ \text{ A} \\ \bar{I}_3 &= 10.37\angle -21.80^\circ \text{ A} \end{aligned}$$

The currents in the outer lines are thus

$$\begin{aligned} \bar{I}_{aA} &= \bar{I}_1 = 11.24\angle -19.83^\circ \text{ A} \\ \bar{I}_{bB} &= -\bar{I}_3 = 10.37\angle 158.20^\circ \text{ A} \end{aligned}$$

The smaller neutral current is

$$\bar{I}_{nN} = \bar{I}_3 - \bar{I}_1 = 0.9459\angle -177.7^\circ \text{ A}$$

The average power drawn by each load may thus be determined:

$$\begin{aligned} P_{50} &= |\bar{I}_1 - \bar{I}_2|^2(50) = 206 \text{ W} \\ P_{100} &= |\bar{I}_3 - \bar{I}_2|^2(100) = 117 \text{ W} \\ P_{20+j10} &= |\bar{I}_2|^2(20) = 1763 \text{ W} \end{aligned}$$

The total load power is 2086 W.

The loss in each of the wires is next found:

$$P_{aA} = |\bar{I}_1|^2(1) = 126 \text{ W}$$

$$P_{bB} = |\bar{I}_3|^2(1) = 108 \text{ W}$$

$$P_{nN} = |\bar{I}_{nN}|^2(3) = 3 \text{ W}$$

Total line loss of 237 W.

5. For the circuit shown in the figure below, find the average power supplied by the source and the average power absorbed by the resistor.

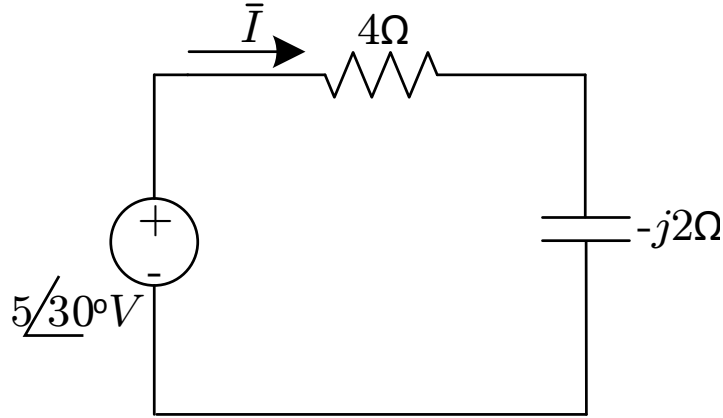


Figure 13

Solution:

The current $\bar{I}$ is given by

$$\bar{I} = \frac{5/30^\circ}{4 - j2} = \frac{5/30^\circ}{4.472/-26.57^\circ} = 1.118/56.57^\circ \text{ A}$$

The average power supplied by the voltage source is

$$P = \frac{1}{2}(5)(1.118)\cos(30^\circ - 56.57^\circ) = 2.5 \text{ W}$$

The current through the resistor is

$$\bar{I}_R = \bar{I} = 1.118/56.57^\circ \text{ A}$$

and the voltage across it is

$$\bar{V}_R = 4\bar{I}_R = 4.472/56.57^\circ \text{ V}$$

The average power absorbed by the resistor is

$$P = \frac{1}{2}(4.472)(1.118) = 2.5 \text{ W}$$

which is the same as the average power supplied. The capacitor absorbs zero average power.

6. A load $\bar{Z}$ draws 12 kVA at a power factor of 0.856 lagging from a 120 V (RMS) sinusoidal source. Calculate (A) the average and reactive powers delivered to the load, (B) the peak current, and (C) the load impedance.

Solution:

Part A

Given that, $\text{pf} = \cos\theta = 0.856$, we obtain the power angle as $\theta = \cos^{-1}0.856 = 31.13^\circ$.

If the apparent power is $S = 12,000 \text{ VA}$, then the average or real power is

$$P = S \cos\theta = 12,000 \times 0.856 = 10.272 \text{ kW}$$

while the reactive power is

$$Q = S \sin\theta = 12,000 \times 0.517 = 6.204 \text{ kVAR}$$

Part B

Since the pf is lagging, the complex power is

$$\bar{S} = P + jQ = 10.272 + j6.204 \text{ kVA}$$

From $\bar{S} = \bar{V}_{rms} \bar{I}_{rms}^*$, we obtain,

$$\bar{I}_{rms}^* = \frac{\bar{S}}{\bar{V}_{rms}} = \frac{10,272 + j6204}{120\angle 0^\circ} = 85.6 + j51.7 \text{ A} = 100\angle 31.13^\circ \text{ A}$$

Thus $\bar{I}_{rms} = 100\angle -31.13^\circ \text{ A}$ and the peak current is

$$I_m = \sqrt{2}I_{rms} = \sqrt{2}(100) = 141.4 \text{ A}$$

Part C

The load impedance

$$\bar{Z} = \frac{\bar{V}_{rms}}{\bar{I}_{rms}} = \frac{120\angle 0^\circ}{100\angle -31.13^\circ} = 1.2\angle 31.13^\circ \Omega$$

which is an inductive impedance.

_____ **That's it.** _____